Program to Compute Quadrupole-Dipole Resorbed Atomic Charges

version 1.0

April 9, 2025

1. Introduction:

This program computes the quadrupole-dipole resorbed (QDR) point charges and the residual atomic dipoles and residual atomic quadrupoles as described in the following journal article: Alma Carolina Escobosa and Thomas A. Manz, "How well do various QM-derived net atomic charges reproduce the electrostatic potential surrounding a material across multiple geometric conformations?", submitted to RSC Advances, 2025.

2. Compiling the program:

To date, this program has only been compiled and tested on Linux operating system. If using the Linux operating system, you may be able to use the pre-compiled executable file named **QDR_program_04_09_2025_parallel** (parallel version) or **QDR_program_04_09_2025_serial** (serial version). Be sure to change permissions on this file to make it executable, then try to run it. If that does not work for your system, then you need to recompile the program from the source files using the gfortran compiler. In the sourcecode subdirectory, change permission on the **compile_parallel.txt** (or **compile_serial.txt**) file to make it executable, then cd into this sourcecode subdirectory and type **./compile_parallel.txt** (or **./compile_parallel.txt**) at the Linux command prompt to compile the program.

Note: For unit cells containing millions of atoms, 8-byte (64-bit) integers are required, because the normal 4-byte integers do not provide sufficient range for the large array indices. For these large systems, you can use the executables **QDR_program_04_09_2025_parallel_int8** (parallel version with 8-byte integers) or **QDR_program_04_09_2025_serial_int8** (serial version with 8-byte integers).

3. Running the program:

Before running this program, you must first run the Chargemol program to analyze your material. Then place the following files into a subdirectory:

- **DDEC_net_atomic_charges.xzy** – This is one of the files that was generated by Chargemol.
- **overlap_populations_and_contact_exchanges.txt** – This is one of the files that was generated by Chargemol.

If using a batch queue on a Linux cluster to run the program, include your batch submission script in the same subdirectory as the above input files. This is not needed if you run the program directly using the command prompt.

To run the program, cd (change directory) into the subdirectory containing the above input files, then use one of the following commands to run the program:

/path_to_program/QDR_program_04_09_2025_parallel

/path_to_program/QDR_program_04_09_2025_serial

where "/path_to_program/" is the directory containing the compiled program executable. The program does not need to be in the same directory as the above input files.

Please note the following: A developmental (pre-release) Chargemol version must be used to generate the required input files. To receive a copy of this developmental (pre-release) Chargemol version, you must send an email request to tmanz@nmsu.edu. (This version will be released through sourceforce in the near future, but has not yet been publicly released.) Earlier versions of the Chargemol program did not generate the* overlap_populations_and_contact_exchanges.txt *file.

4. Included examples: Please see the subfolder named **worked_examples** containing examples of input and output files. The program produces two output files:

QDR-DDEC6_point_charges.xyz – This file contains the computed QDR atomic charges, residual atomic dipoles, and residual atomic quadrupoles. For molecules, the overall molecular dipole and traceless quadrupole moments (computed from the QDR point charges, and/or atomic dipoles, and/or atomic quadrupoles) are printed at the bottom of the file.

QDR_charges_computation_logfile.txt – This is a log file that prints some stuff regarding progress of the calculation.

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